

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458042.7015 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1184 +/- 0.0059
Transit depth (Rp/Rs)^2	1.4 +/- 0.14 [%]
Orbital Inclination (inc)	88.62 +/- 1.23
Airmass coefficient 1 (a1)	1.0153 +/- 0.0077
Airmass coefficient 2 (a2)	-0.0143 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#2 - [356, 153]
Optimal Aperture	4.36
Optimal Annulus	9.87
Transit Duration (day)	0.1546 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1184 $\pm$ 0.0059 vs 0.1036 (deviation 2.15 σ)
Duration fit vs published	0.1546 d vs 0.1567 d (deviation 0.68 σ)
Mid-time O-C	-8.4 min
Depth SNR	17.2
Residual scatter	0.76 %

Light curve

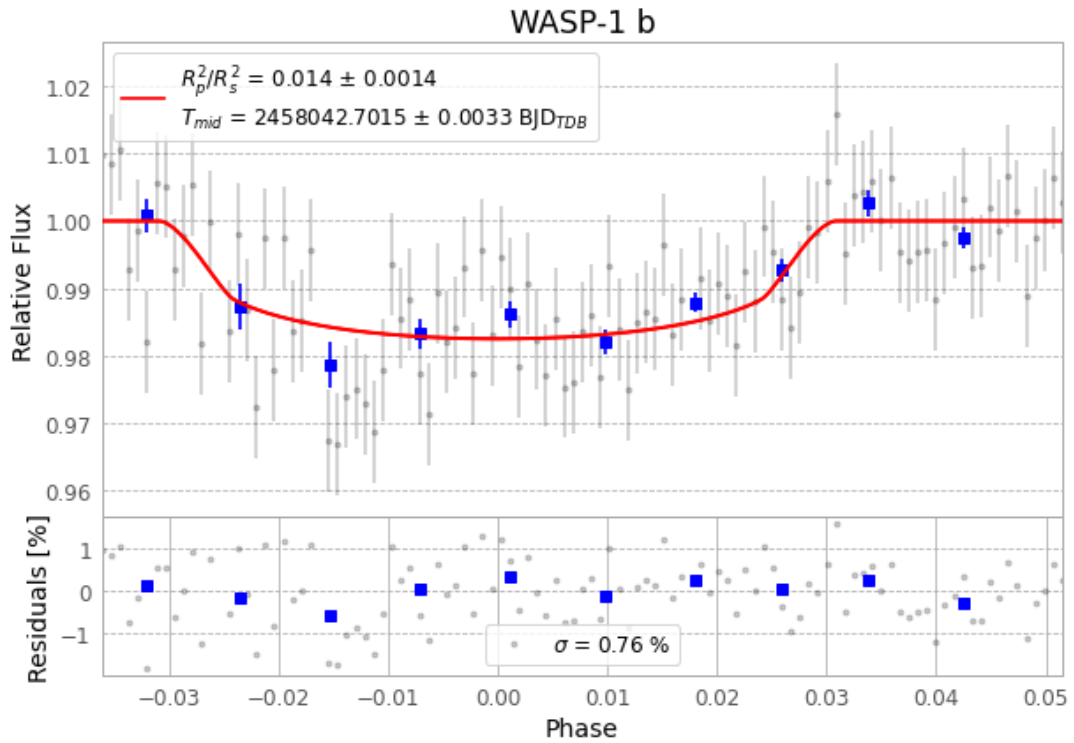


Figure 1 — FinalLightCurve_WASP-1 b_2017-10-15.png (SHA-256 eb8f7f22b17d8ce2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0b76a00557979ddb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-1171016023012.FITS → WASP-1171016074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-171016.log (4e86c8b98761...), FinalLightCurve_WASP-1 b_2017-10-15.png (eb8f7f22b17d...), FinalLightCurve_WASP-1 b_2017-10-15.csv (2135113e3979...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:41Z.